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Model Question

Unit-2: Modulation, Transmission Media and Switching Techniques

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Subject: Data Communication & Networks

Subject Code: ECE21302

PART- A (2 MARKS)

1. Illustrate the requirement of analog-to-analog modulation techniques.
2. Discussed the requirement of analog to digital conversion approaches.
3. Explain the need of data encoding before sending over a medium?
4. What are the different the possible analog-to-analog modulation techniques?
5. Compare the bandwidth requirement of the three analog-to-analog modulation techniques?
6. Between AM and FM, which one gives better noise immunity?
7. What are the possible digital-to-analog modulation techniques?
8. Why PSK is preferred as the modulation technique in modems?
9. Out of the three digital-to-analog modulation techniques, which one requires higher bandwidth?
10. How do guided media differ from unguided media?
11. Write the significance of the twisting in twisted-pair cable?
12. What are the three major classes of guided media?
13. Name the advantages of optical fiber over twisted-pair and coaxial cable.
14. Explain the purpose of cladding in an optical fiber?
15. In what situation multiplexing is used?
16. Distinguish between the two basic multiplexing techniques?
17. What is guard bands? Write the importance of the guard bands in case of FDM?
Write the value of it for AM and FM transmission.
18. What are the possible encoding techniques? Give examples.
19. What are the differences between parallel and serial transmission?

PART- B (7 MARKS)

1. What is transmission medium? What are the different types of transmission medium?
Illustrate about Guided and Un-guided medium.
2. What do you mean by switched network? Write down the requirements of switching network? Compare and contrast a circuit switched network and packet switched network with suitable diagrams.
3. Describe the operation of circuit switched network and packet switched network with suitable diagrams. Summarize the difference between circuit switched network and packet switched network.
4. Explain the key features of FDM and TDM. Distinguish between synchronous and asynchronous TDM.

5. What is packet switching? Discuss two different types of packets switching.
6. Illustrate about the different transmission media? Differentiate between guided media and unguided media. Mention the advantages of optical fiber over twisted pair and coaxial cable.
7. Define carrier signal and its role in analog transmission of digital data. What are the different analog and digital data modulation schemes? Discussed any one analog and digital modulation scheme.
8. What is Multiplexing and Explain different types of Multiplexing?